

Docket Number
IMY-2023-15495
Safety Measurement
in Public Environments
Using IoT Technology
Final Report from the Swedish Authority for Privacy Protection
Regulatory Sandbox on Data Protection

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Using IoT Technology – Final Report
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Table of Contents

Summary	3
1.	
Introduction	5
1.1	
Regulatory Sandbox – a Form of Experimental Activity	5
1.2	
IMY's First Pilot Project	7
1.3	
Permanent Part of IMY's Operations and Name Change	7
1.4	
Selection and Work Process in the Second Pilot Project	7
2.	
The Project "Safety Measurement in Public Environments Using IoT Technology"	9
2.1 Brief About the Participants	9
2.2 Brief About the Current Pilot Project	9
2.3 The Traffic Office's Purpose with the Intended Processing	9
2.4 What is the Internet of Things (IoT)?	11
2.5 Safety-Enhancing Measures	11
2.6 The Legal Questions in the Pilot Project	12
2.7 Limitations in the Pilot Project and Other Information	12
3.	
Description of the Technology in the Current Project	13
3.1	
LiDAR	13
3.1.1 Brief About Edge Calculation	14
3.1.2 Brief About the AI Technology Intended to Be Used	14
3.2	
Technical Specifications	15
4.	
Can Data Collected via LiDAR Sensors Constitute	

Personal Data Processing According to the Data Protection Regulation?	16
4.1	
Personal Data	16
4.1.1 What is Personal Data?	16
4.1.2 IMY's Comments	16
Docket Number:	
IMY-2023-15495	
Date:	
2024-02-08	
Swedish Authority for Privacy Protection	
Docket Number: IMY-2023-15495	
2(28)	
Date: 2024-02-08	
4.2	
Processing of Personal Data	17
4.2.1 What is Processing?	17
4.2.2 IMY's Comments	17
4.3	
Sensitive Personal Data	17
4.3.1 What is Sensitive Personal Data?	17
4.3.2 IMY's Comments	17
4.4	
Possible Avenues for Further Investigation	17
5.	
How Does the Use of LiDAR Sensors Relate to	
The Camera Surveillance Act?	19
5.1	
Is LiDAR Technology Covered by the Concept of "Camera Surveillance" in	
The Camera Surveillance Act?	19
5.1.1 TV Camera, Other Optical-Electronic Instrument or Comparable Equipment	
.....	19
5.1.2 IMY's Comments	20
5.2	
Does it Involve Permanent or Regularly Repeated	
Personal Surveillance?	21
5.2.1 Permanent or Regularly Repeated Personal Surveillance	21
5.2.2 IMY's Comments	21
5.2.3 Possible Avenues for Further Investigation	22
5.3 Surveillance Requiring Permission – What Considerations Need to Be Made	
and How is the Assessment Affected by New Technology?	23
5.3.1 Is the Use of LiDAR Sensors in a Public Place Subject to Permission?	
.....	23
5.3.2 Purposes to Be Particularly Considered in a Permission Review	23
5.3.3 IMY's Comments	24
5.4 The Camera Surveillance Act is Under Review	25
6.	
Reflections on the Regulatory Sandbox as a Working Method	26
7.	
Reflections on Applying Data Protection Regulations to New Technology ...	28
Swedish Authority for Privacy Protection	
Diary Number: IMY-2023-15495	

Date: 2024-02-08

Summary

- Complex societal challenges require innovative solutions. Sweden is facing significant societal challenges, including climate change and demographic changes, where new and existing technology will often be viewed as key tools to address these challenges. Artificial intelligence (AI), the Internet of Things (IoT), and other innovative technologies provide new tools to contribute to safety, health, welfare, and growth.
- The pace of technological development challenges the ways society operates. To manage the timing issue between rapid technological development and slow-moving regulatory processes, the use of regulatory sandboxes can be an important approach. There is no widely accepted definition of what this approach entails, but the core idea is that innovators and, for example, regulatory authorities work together to interpret and apply regulations to innovative products and services. These regulatory sandboxes aim to increase legal predictability, promote innovation, and facilitate conditions for, for example, startups and small businesses.
- The Swedish Authority for Privacy Protection (IMY) has conducted its second pilot project within the authority's regulatory sandbox during the summer and autumn of 2023. By regulatory sandbox, IMY refers to enhanced guidance on how data protection regulations should be interpreted and applied. Characteristic of this approach is that IMY, together with the relevant business, identifies the legal issues that the guidance will focus on. Guidance is then provided orally on several occasions over a few months in the form of workshops or other dialogue-based formats. The work culminates in a public report where discussions and assessments are summarized to enable learning for a broader audience.
- The project "Safety Measurement in Public Spaces Using IoT Technology" is IMY's second pilot project. The participants in this project have been the Traffic Office of the City of Stockholm, Internet of Things Sweden, and Kista Science City AB. The project concerns the Traffic Office's desire to use LiDAR sensors to systematically and uniformly gather data on the proportion of women, men, and children moving in a public square. Safety surveys show that women avoid places they perceive as unsafe. Hence, the hypothesis is that places where women and children are present are perceived as safer than others. By obtaining a representative overview of which groups are present in a specific location, the Traffic Office can plan and take safety-enhancing measures to promote the presence of women and children if they are underrepresented in that location.
- LiDAR sensors are a form of IoT technology. A LiDAR sensor emits infrared laser pulses to measure the distance to an object and then creates a three-dimensional point cloud. The more light points that hit an object within the sensor's detection range, the more detailed point clouds can be obtained regarding that object. Within the framework of the pilot project, the output generated by a LiDAR sensor is intended to serve as input to an AI algorithm. This algorithm will then be able to classify whether an individual within the LiDAR sensor's detection range is a woman, a man, or a child.
- The guidance in the pilot project has focused on two legal questions. In addition to these two questions, there are other legal issues that need to be considered but have not been analyzed within the framework of the project.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

Date: 2024-02-08

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Question 1. Can data collected via LiDAR sensors constitute personal data processing under the General Data Protection Regulation? IMY finds that there is much to suggest that there are identifiers in the output generated by a LiDAR sensor and that this data could constitute personal data under the General Data Protection Regulation. IMY further finds that the process that occurs when a LiDAR sensor generates output typically constitutes processing under the General Data Protection Regulation. Likewise, there is much to suggest that personal data processing also occurs when the AI algorithm performs calculations based on the output from the LiDAR sensor.

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Question 2. How does the use of LiDAR sensors relate to the Camera Surveillance Act? IMY finds that LiDAR sensors typically fall under the concept of "other optical-electronic instruments" in the Camera Surveillance Act. There is much to suggest that it may be possible to discern and identify individuals based on, for example, body movements, body constitution, and clothing in the output generated by a LiDAR sensor. If surveillance occurs in a public place and the actor is an authority, it thus appears likely that it would constitute such repeated personal surveillance that falls within the scope of the Camera Surveillance Act. IMY notes that the use of LiDAR sensors for the purpose relevant to the project is not obviously a purpose that should be particularly considered under the Camera Surveillance Act when assessing the surveillance interest. However, IMY finds that the technology can help limit privacy intrusion, especially in comparison with traditional surveillance cameras.

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IMY has now completed two pilot projects in its regulatory sandbox and decided to make the working method a permanent part of its operations. In 2024, IMY will conduct new projects in both the spring and autumn. Experiences show that this working method is beneficial for both innovation actors and supervisory authorities, but also for society at large by promoting the development of new technology while contributing to ensuring data protection. Many public organizations have contacted IMY to learn more about the regulatory sandbox. IMY has also collaborated and shared experiences with other data protection authorities in Europe. According to the upcoming AI Regulation, member states are required to establish regulatory sandboxes for AI, and IMY is expected to be involved as the national data protection authority. Despite differences in regulation, IMY's experiences will be valuable.

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Cross-functional work is crucial for successfully driving innovation while ensuring good data protection. Based on the pilot project, IMY makes some general reflections on the cross-functional working method needed in innovation processes. An important experience is that, in order to make relevant legal assessments, a relatively deep understanding of the technology is required. Having good pedagogical skills is therefore necessary for both technicians and lawyers. Structures and tools to ensure a common understanding can facilitate the work. It may also be advantageous to continuously revisit and, if necessary, adjust or supplement the legal questions that need to be investigated.

Integrity Protection Authority

Diary Number: IMY-2023-15495

5(28)

Date: 2024-02-08

1. Introduction

Sweden faces significant societal challenges in a number of areas. Climate change, security threats, energy supply, and demographic changes are examples of phenomena that have fundamentally changed, and will continue to change, the conditions for welfare.

Common to the societal challenges of our time is that new technology, or existing technology used in new ways or on a new scale, will often be part of the solution. Technological development is advancing rapidly and constantly offers new opportunities. Artificial intelligence (AI), the Internet of Things (IoT), quantum computers, and other innovative technologies provide new tools to contribute to security,

health, welfare, and growth.

In Sweden, there is an increasingly intense discussion about how the public administration can meet complex societal challenges and fully leverage new or existing technology. The pace of technological development challenges society's familiar working methods, structures, leadership, roles, and regulations. In international literature, the term "pacing problem" is often used to describe the difference in speed between exponential technological development and the often sluggish regulatory process. In Sweden, for example, the authority Growth Analysis has highlighted the challenges of regulating technological innovation.¹

In the innovation system, there is a strong desire to do the right thing and to develop sustainable solutions, but at the same time, there is genuine uncertainty about how regulations should be applied to new technology. There is a risk that the precautionary principle may take precedence and hinder innovation. For supervisory authorities such as the Integrity Protection Authority (IMY), one of the main challenges is to keep up with understanding and analyzing the new technology to which existing regulations will be applied.

1.1 Regulatory Sandbox – a Form of Experimental Activity

Among other things, the OECD believes that an important part of the way forward involves strengthening the innovation capacity in the public sector.² In Sweden, the Committee for Technological Innovation and Ethics (Komet) has been a strong voice advocating for an acceleration of the pace of change in the public sector so that Sweden can fully benefit from the opportunities of digitalization. One of the development paths proposed by Komet is to drive experimental activities to a greater extent.³

A regulatory sandbox is a specific form of experimental activity where developers and, for example, supervisory authorities work together to interpret how the regulations can be applied in practice to innovative products and services. The intention is, among other things, to increase legal predictability, shorten the time for an actor's product or service to reach the market, and facilitate for startups and small businesses. Regulatory sandboxes are sometimes also referred to as sandboxes, test beds, greenhouses, or incubators.

Within the EU, among other things, the Council of Ministers has highlighted regulatory sandboxes as a way to contribute to innovation and growth for businesses.⁴ The recently agreed AI-

1 Challenges in Regulating Technological Innovation – Possible Policy Measures, Growth Analysis Report 2022:04.

2 OECD Declaration on Public Sector Innovation, OECD/LEGAL/0450.

3 Renewing Tactics in Line with Technology – Proposals for a Responsible, Innovative, and Collaborative Administration, SOU 2022:68.

4 Council of the European Union. Adopted Council Conclusions on Regulatory Sandboxes and Experimentation Clauses as Tools for an Innovation-Friendly, Future-Proof, and Resilient Regulatory Framework Addressing Disruptive Challenges in a Digital Age. Brussels, November 16, 2020. The Council defined regulatory sandboxes as a structured...

Swedish Authority for Privacy Protection

Case Number: IMY-2023-15495

6(28)

Date: 2024-02-08

The regulation will also contain provisions regarding the establishment of regulatory sandboxes as a means to promote and facilitate the development of innovative AI systems.⁵

In the area of data protection, IMY's sister authorities in the United Kingdom, Norway, and France have conducted regulatory sandboxes in recent years, providing guidance on the application of the General Data Protection Regulation⁶, commonly abbreviated as GDPR. For these data protection authorities, there has been no specific regulation for the trial activities or any exceptions from the applicable data protection framework.

The fundamental idea behind a regulatory sandbox in the area of data protection is that the supervisory authority provides exploratory, dialogue-based guidance to innovation actors in exchange

for the work being summarized in a public report that enables learning for others. Thus, practical examples are developed in gray area issues, that is, issues where practice or other guidance is lacking, in areas where both the technology and the law are complicated, relatively new, and untested. At the same time, this working method contributes to increasing the supervisory authority's understanding of new technology and how it can be applied. The ability of supervisory authorities to keep up with technological developments and understand how new technology works is a prerequisite for them to subsequently apply the law in an adequate and appropriate manner.

Context for experiments that allow for testing innovative technologies, products, services, or methods (currently particularly in connection with digitalization) in a real environment for a limited period and in a limited part of a sector or area under governmental supervision to ensure that appropriate safeguards are in place.

5 <https://www.consilium.europa.eu/en/press/press-releases/2023/12/09/artificial-intelligence-act-council-and-parliament-strike-a-deal-on-the-first-worldwide-rules-for-ai/>

6 Regulation (EU) 2016/679 of the European Parliament and of the Council of April 27, 2016, on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).

By regulatory sandbox, IMY refers to in-depth guidance on how the applicable data protection framework should be interpreted and applied. No exceptions are granted from the rules in the data protection regulation or its supplementary legislation.

Characteristic of the working method is that:

- IMY and innovation actors together identify the legal issues that the guidance will focus on,
- guidance is provided orally on several occasions over a few months in the form of workshops or other dialogue-based formats, and
- reasoning and assessments from the guidance are summarized and published in a report to enable learning for many.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

7(28)

Date: 2024-02-08

1.2 IMY's First Pilot Project

In March 2023, the first pilot project in IMY's regulatory sandbox was concluded. The participants in the project were the healthcare providers Region Halland and Sahlgrenska University Hospital. AI Sweden, which is Sweden's national center for applied AI, was also involved and provided its extensive knowledge in artificial intelligence. The theme of the project was decentralized AI in healthcare. The healthcare providers wanted to evaluate the possibilities of jointly training and exchanging machine learning models using federated machine learning. IMY's reasoning and assessments regarding the project are summarized in the report Federated Machine Learning Between Two Healthcare Providers (dnr IMY-2023-2602). The report is published on IMY's website.⁷

1.3 Permanent Part of IMY's Operations and Name Change

More and more innovators seek guidance on data protection to embed strong data protection in their products or services from the outset. By operating a regulatory sandbox, IMY can address gray area issues at an earlier stage and in a different manner than through, for example, the authority's supervisory activities. The experiences from IMY's pilot project have also been very positive. From 2024, guidance through the regulatory sandbox has therefore become a permanent part of IMY's operations. During the year, IMY will conduct two new projects: one starting in the spring and one starting in the autumn.

During the first pilot project, IMY used the term regulatory testing activity. IMY has since chosen to change the name to regulatory sandbox as this is a term that is more established within the innovation system. The latter is also used by IMY's sister authorities in, for example, Norway and the United Kingdom for their corresponding trial activities.⁸ However, the name change does not mean that IMY has changed its working methods; they remain the same as since the first pilot.

1.4 Selection and Work Process in the Second Pilot Project

The selection of innovation projects for the second pilot took place after various organizations submitted unsolicited inquiries to IMY, such as the Medical Products Agency, Internet of Things Sweden (IoT Sweden), and Sjyst data! (a collaboration and innovation project on data protection and privacy run by RISE).

To participate in the pilot project, the following criteria needed to be met:

- The project was at such an early stage that the intended personal data processing had not yet commenced,
- The project had progressed sufficiently to define relatively concrete legal questions,
- The project matched IMY's planning in time and could allocate resources for a number of workshops during the summer and autumn of 2023, and
- The relevant organizations accepted that IMY's pilot project would result in a public final report.

Most of the projects presented to IMY met the above criteria. One of the main reasons IMY conducts a regulatory sandbox is for the authority to acquire knowledge about new technology more quickly,

7 <https://www.imy.se/publikationer/slutrapport-om-imys-pilotprojekt-med-regulatorisk-testverksamhet-om-dataskydd/>

8 The Data Protection Authority in Norway uses the term sandkassen (Swedish: sandlådan) and the Information Commissioner's Office (ICO) in the United Kingdom uses the term regulatory sandbox (Swedish: regulatorisk sandlåda).

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

8(28)

Date: 2024-02-08

especially in areas where technology has gained, or will soon gain, significant importance in society. The Internet of Things (IoT) is one of the technology areas where much has happened in a short time, and something that IMY highlighted as a particular risk area in the Privacy Protection Report 2020 (Report Number: 2021:1). IoT is also one of the technology areas where the number of granted patents has increased the most in recent years. The use of various IoT products in society has also increased significantly. It was against this background that IMY chose one of the projects from IoT Sweden's portfolio to become the second pilot project in IMY's regulatory sandbox. The theme of the current project is safety measurement in public environments using IoT technology. The participants in the pilot project have been IoT Sweden, Kista Science City AB, and the Traffic Office in the City of Stockholm.

The practical work in the pilot project has consisted of three phases:

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Start-up phase. In June 2023, initial meetings were held to understand the current innovation project and the expectations and conditions with which the various participants entered the regulatory sandbox. IMY and the participants selected the questions that the guidance would focus on. The start-up phase also included a joint press release and joint activities during Almedalsveckan.

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Guidance phase. The main part of the work was carried out in the form of a series of workshops during the period of August–October 2023. Dialogue was held regarding the legal questions, and IMY provided ongoing verbal guidance. A couple of special meeting occasions were dedicated to in-depth examinations of the relevant technology. At the concluding meeting, IMY and the participants also reflected together on the working method with the regulatory sandbox, what worked well, and what could be developed. A total of about 15 people from IMY, IoT Sweden, Kista Science City, and the Traffic Office participated in the work.

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Reporting phase. During November 2023–January 2024, IMY worked on compiling this final report, which describes the work and the legal considerations from the pilot project. IoT Sweden, Kista Science City, and the Traffic Office have helped review the report.

9 Privacy Protection Report 2020 p. 61 (<https://patentscope.wipo.int/search/en/search.jsf> [the search was conducted with the setting "Any Field"]).

10 <https://www.imy.se/nyheter/trygghetsprojekt-vagleds-i-grazonsfragor-om-dataskydd/>

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

9(28)

Date: 2024-02-08

2. Project "Safety Measurement in Public Environments Using IoT Technology"

2.1 Brief about the Participants

The Internet of Things Sweden (IoT Sweden) is a strategic innovation program aimed at creating societal benefits through increased use of the Internet of Things in the public sector. IoT Sweden supports and finances innovative IoT projects within the public sector across the country. IoT Sweden is coordinated by Uppsala University, and the program is jointly financed by Vinnova, the Energy Agency, and Formas.

Kista Science City AB is a non-profit organization that works on the development of new technology and serves as an innovation arena where business, academia, and society meet to jointly drive development forward. Through collaboration, programs, and projects, Kista Science City works on the development of companies and skills supply. Kista Science City is a subsidiary of the Electrum Foundation.

The Traffic Office in the City of Stockholm (Traffic Office) is responsible for ensuring that public spaces within the municipality are safe and attractive with effective transport solutions and good accessibility. The elected representatives in Stockholm City's Traffic Committee decide how the municipality should work in the various areas. The Traffic Office is then responsible for implementing the Traffic Committee's decisions.

2.2 Brief about the Current Pilot Project

The pilot project concerns the Traffic Office's desire to use so-called LiDAR sensors to collect data on the proportion of women, men, and children moving in Jan Stenbeck's Square in the Kista district. The Traffic Office's ambition is to obtain a situational picture of how the population mix, that is, the distribution between the number of men, women, and children, looks at different times at the current location through an automated approach.

Currently, the Traffic Office has not installed any LiDAR sensors in Jan Stenbeck's Square, and it is also not certain that this will happen. Through IMY's pilot project, the Traffic Office has received guidance on whether there are conditions under the applicable data protection regulations to proceed with these plans.

2.3 The Traffic Office's Purpose with the Intended Processing

One of the first data protection legal questions that needs to be answered in all innovation processes is what purpose the intended personal data processing has. The Traffic Office has described the purpose as follows:

To create attractive, vibrant, and safe public places, the municipality needs to repeatedly measure the indicators that characterize such places over time. This is necessary to know which places are attractive, vibrant, and safe, but also to know which are not, in order to direct measures and follow up with initiatives.

11 Read more in section 3.1 below.

12 In this context, children refer to individuals up to 12 years old, according to the Traffic Office.

13 Read more on IMY's website (<https://www.imy.se/verksamhet/dataskydd/innovationsportalen/ar-det-ni-tankt-gora-ar-forenligt-med-gdpr/>).

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

10(28)

Date: 2024-02-08

Measuring the composition of people, specifically the proportion of women and children, present in a certain location provides a good indication of whether the place is attractive and safe for everyone. The Traffic Office has stated that the purpose of the data collection is not to prevent or investigate crimes at the location, but rather to examine if and when women and children use the area. Safety surveys show that women avoid places they perceive as unsafe; hence, the hypothesis is that locations where women and children are present are perceived as safer than others. This has led many cities in the Western world to measure the proportion of women and children at various times as an indicator of how safe or unsafe a place is perceived to be. By obtaining a representative overview of which groups are present in a certain location, the Traffic Office can plan and implement safety-enhancing measures to promote the presence of women and children if they are underrepresented in that area.

The Traffic Office has produced situational overviews of the demographic mix in various public places on three occasions (1995, 1996, and 2022). The Traffic Office has also conducted a number of measurements on an ad hoc basis. Such measurements have been carried out entirely manually by sending out consultants, students, or employees to count on-site. The Traffic Office has stated that this manual method is resource-intensive and costly. The ambition of the Traffic Office is to measure the demographic mix in a more uniform and systematic way. By conducting measurements before and after various interventions, as well as monitoring how selected locations are utilized by residents over time, the Traffic Office could take targeted and appropriate actions to create a safer urban environment. The goal within the framework of this pilot project is to conduct measurements at Jan Stenbeck Square during certain time intervals between 07:00 and 22:00 two days per month.

Image 1. Jan Stenbeck Square in Kista (Stockholm Municipality).

14 Crime Prevention Council 2023: National Safety Survey; Listerborn, C. 2002: Safe City; discourses on women's fear in research, policy development, and local practice. (Doctoral dissertation, Chalmers University of Technology, Gothenburg).

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

11(28)

Date: 2024-02-08

2.4 What is the Internet of Things (IoT)?

The Internet of Things, or Internet of Things (IoT), is a broad concept typically used to describe networks of smart devices that sense or interact with their environment. The devices in the network can continuously collect information, respond to it, and communicate both with people and with other devices. Everyday objects such as appliances, speakers, thermostats, lighting, televisions, electronic locks and alarms, clothing, or cars, as well as equipment and sensors in industry, infrastructure, or healthcare can contain IoT technology. The trend is moving towards IoT being used in an increasing number of societal areas and in more geographical locations to collect data and streamline workflows, among other things.

Within the framework of "smart cities," connected devices and sensors in public spaces are becoming increasingly common. The underlying purposes of collecting data may, for example, be to streamline and automate various types of controls and measurements. Examples of applications can be found in the transportation sector, where it may involve connecting vehicles with smartphones and other vehicles. IoT sensors are also used in road and railway networks and in public transport for increased traffic safety, vehicle diagnostics, and more. In properties, sensors can, for example, collect data for more efficient management of security, lighting, and diagnostics. Connected devices can also be found in measuring equipment in smart grids and in transport and water systems. By being connected to a network, the devices can be remotely controlled by people or by other machines. They can also send data used for measurements, diagnostics, and automatic control.

2.5 Safety-Enhancing Measures

The Traffic Office wants to measure the population mix at the relevant location and, using this information, design appropriate safety-enhancing measures aimed at leveling out any disparities

between different groups. In other words, the purpose is not to prevent crime. In this context, the Act (2023:196) on municipalities' responsibility for crime prevention work can be mentioned. The law came into force on July 1, 2023, and aims to ensure that municipalities, based on a knowledge-based foundation, assess the need for crime prevention measures and also take certain responsibility for the coordination of crime prevention work within the municipality. According to Section 4, municipalities must produce a situational picture of crime within the municipality's geographical area. The situational picture must, among other things, include a mapping of crime and other relevant information in this context. Section 5 states that municipalities, based on the situational picture, must decide on a plan for what measures the municipality intends to take to prevent crime (action plan).

The concept of crime prevention measures is not defined in the law on municipalities' responsibility for crime prevention work. It is a difficult-to-define concept that lacks a generally accepted definition.

According to the preparatory works for the law, the concept may also encompass certain safety-enhancing measures. However, it is unclear to what extent. For example, it is not given whether the Traffic Office's intended data collection regarding the population mix at a certain public place can be considered a measure covered by the law. The same applies to the safety-enhancing measures that the city would take to reduce any disparity between women, men, and children at the location.

15 <https://iotsverige.se/iot-sa-funkar-det>

16 See Section 3 of the Act on municipalities' responsibility for crime prevention work.

17 Prop. 2022/23:43 p. 15.

18 Prop. 2022/23:43 p. 15.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

12(28)

Date: 2024-02-08

2.6 The Legal Questions in the Pilot Project

IMY, IoT Sweden, Kista Science City, and the Traffic Office agreed to focus on two legal questions in the current pilot project:

1. Can data collected via LiDAR sensors constitute personal data processing under the General Data Protection Regulation?

2. How does the use of LiDAR sensors relate to the Camera Surveillance Act (2018:1200)?

Sections 4–5 below provide an account regarding the two legal questions.

2.7 Limitations in the Pilot Project and Other Information

In projects like this, a range of different data protection legal questions arise that an organization needs to analyze. Within the framework of this pilot project, there has not been room to address all legal questions. The questions that have been selected and investigated have been deemed highly relevant for the progress of the current project, but also relevant for other actors such as other municipalities considering collecting data in a manner similar to what the Traffic Office wishes to do. It should be emphasized that it is the responsibility of each data controller to ensure that its personal data processing is compliant with the General Data Protection Regulation and supplementary national law.

In this pilot project, IMY has analyzed LiDAR technology. The assessments that IMY has made, and which are detailed below, cannot necessarily be directly applied to other technologies such as radar technology, light sensors, or motion sensors.

This final report contains an account of IMY's views on legal issues where there is a lack of guiding case law or guidance from the European Data Protection Board (EDPB). The assessments are made against the backdrop of the current legal situation and may change if new legislation, case law, or guidance from the EDPB emerges.

19 <https://www.imy.se/verksamhet/dataskydd/dataskydd-pa-eu-niva/edbp/>

Swedish Data Protection Authority

Reference Number: IMY-2023-15495

Date: 2024-02-08

3. Description of the technology in the current project

3.1 LiDAR

To achieve the purpose of the project, namely to measure the population mix at the relevant location, the project participants wish to understand whether LiDAR sensors are a better option from a privacy perspective compared to traditional cameras. LiDAR is an abbreviation of the English term light detection and ranging and is sometimes also referred to as LADAR or laser distance measurement. LiDAR technology has long been used in landscape modeling and in the construction and building industry. Today, it is also used in autonomous vehicles and vehicles with advanced safety systems. Many mobile phones have built-in LiDAR technology in various forms, including a similar technology used for facial recognition. LiDAR is closely related to other technologies, primarily radar (radio detecting and ranging) and sonar (sound navigation and ranging). However, LiDAR is based on pulses of laser light instead of radio or sound waves.

On a general level, LiDAR works as follows. The LiDAR sensor calculates the distance to an object (such as a person, a sign, or a tree) by emitting infrared laser pulses (the light source) and then measuring the time it takes for the pulses to be reflected back to the light receiver in the sensor. The LiDAR sensor emits these laser pulses at very short time intervals while rotating or tilting the light source in a conical pattern to create a so-called point cloud. The laser pulses consist of invisible light in wavelengths around 900 nanometers²⁰ (nm). How a LiDAR sensor works on a broad level is illustrated by the image below.

Image 2. A simplified illustration of how a LiDAR sensor works. The sensor emits laser pulses into its surroundings. When a laser pulse hits an object (for example, a person or a tree), it is reflected back to the light receiver in the sensor.

²⁰ <https://it-ord.idg.se/ord/nanometer/>

Swedish Data Protection Authority

Reference Number: IMY-2023-15495

Date: 2024-02-08

LiDAR sensors typically generate output²¹ in the form of three-dimensional point maps. Depending on the resolution of the sensor, it can create a very accurate 3D model of both the environment and the objects at a location. When discussing a LiDAR sensor's resolution, it primarily refers to the distance between each light point, but also the maximum distance at which the sensor can detect an object. Regarding the distance between each light point, it is measured in the number of degrees between each point (both vertically and horizontally). Common resolutions today range from 0.1 to 0.4 degrees. A lower number indicates a shorter distance between each light point, resulting in higher resolution and better detail. The sensor's field of view is also measured in degrees (both vertically and horizontally). The choice of which LiDAR sensor to use is therefore of great significance. In short, this means that the more light points that hit an object within the sensor's detection range, the more detailed point clouds (output) can be obtained regarding that object. The same applies to the object's placement in relation to the sensor – the closer the object is, the more light points hit the object. A LiDAR sensor has a maximum distance for detection range (detection range) at which the sensor can detect and measure objects. The maximum distance of the detection range varies between different types of LiDAR sensors.

The sensor's update frequency, that is, the number of points collected by a LiDAR system over a given time period, is measured in hertz (Hz). A higher update frequency means that more points are collected per second, resulting in a denser and more granular point cloud. A higher update frequency means that a larger amount of data is collected per unit of time.

To assess whether individuals can be considered identifiable from a legal perspective, the following characteristics of a LiDAR sensor are particularly crucial:

1. resolution in terms of the number of light points hitting an object,
2. distance to the object and the sensor's detection range,
3. the sensor's update frequency, and
4. the sensor's field of view.

3.1.1 Brief on edge computing

Edge computing refers to the transfer of computing power to the edge of the network, that is, to edge nodes or sensors. This means that data does not need to be sent through the network to a central server where the computation or processing of the current data then occurs. Edge computing allows for certain, and sometimes demanding, data processing to be done directly in or in close proximity to the sensors.

In the current pilot project, the participants intend to use edge-based technology. The processing of the data generated by the LiDAR sensors is thus intended to occur directly in or in close proximity to the sensors.

3.1.2 Brief on the AI technology intended to be used

The output generated by a LiDAR sensor is to serve as input to an AI algorithm. This algorithm will then be able to classify whether an individual, who is within the detection range of the LiDAR sensor, is a woman, a man, or a child. With the help of these

21 <https://it-ord.idg.se/ord/utdata/>

22 <https://it-ord.idg.se/ord/kantdatorsystem/>

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

15(28)

Date: 2024-02-08

Data can enable the Traffic Office to create a situational picture of the population mix in the relevant square.

The AI algorithm is not yet on the market, but according to the participants, it is technically possible to develop such a model. For the algorithm to function appropriately, the output generated by the LiDAR sensors needs to be stored for a certain period to avoid excessive double counting of individuals present at the location. It should be emphasized that the data protection legal issues arising in connection with the development of the algorithm in question are outside the scope of this project. It is not intended that any of the project participants will develop the algorithm themselves; rather, this will be done through external suppliers in the market.

3.2 Technical Specifications

In the project, the participants intend to use 1–3 edge-based LiDAR sensors with a resolution of 0.1 degrees (both horizontally and vertically). The sensors are intended to be placed at a height of 4–8 meters above ground level and have a maximum detection range of 250 meters. The field of view is 115x25 degrees (HxV). The output generated by a sensor with the specified resolution is illustrated below in the form of a 2D figure. It should be noted that the output from a sensor practically forms a moving 3D cloud where depth data is also recorded. The AI algorithm will then classify whether an individual captured by the sensor is a woman, a man, or a child. This data is then sent to the Traffic Office, which will receive data on the population mix at the relevant location. The algorithm will perform the classification, that is, the data processing, directly in or in close proximity to the sensor.

Image 3. InnovizOne Automotive grade LiDAR. Excerpt from video material illustrating the output generated by the selected LiDAR sensor with a resolution of 0.1x0.1 degrees.

In summary, the use of the technology described above is intended to occur in the following steps.

1. The LiDAR sensor emits infrared laser pulses towards objects located up to 250 meters away. The laser pulses are then reflected back to the sensor.
2. The LiDAR sensor calculates the distance to the object and generates output in the form of a 3D model of the environment.

3. The edge-based calculation with the AI algorithm occurs directly in or in close proximity to the LiDAR sensor. The AI algorithm classifies the individuals within the sensor's detection range into one of the groups: woman, man, or child.

4. The output from the AI algorithm provides statistics on the number of women, men, and children. This statistics is then sent to the Traffic Office over the internet.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

16(28)

Date: 2024-02-08

4. Can data collected via LiDAR sensors constitute personal data processing under the General Data Protection Regulation?

4.1 Personal Data

4.1.1 What is personal data?

Article 4.1 of the General Data Protection Regulation defines the concept of personal data. According to the regulation, personal data is any information relating to an identified or identifiable natural person. An identifiable natural person is one who can be identified, directly or indirectly, by reference to an identifier such as a name, an identification number, location data, an online identifier, or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of that natural person.

Personal data can be divided into two subcategories: direct and indirect personal data. A direct personal data can be a name or a personal identification number. An indirect personal data, on the other hand, can be almost anything that can be linked to a natural person in some way (also referred to as identifiers). Recital 26 of the General Data Protection Regulation states that, in assessing whether a person is identifiable, all means that could reasonably be used by the data controller or by any other person to identify the individual, directly or indirectly, should be taken into account. This means that the data controller does not need to have all the information that could make identification possible. However, there must be a reasonable likelihood that the data controller or someone else will be able to identify the individual.

When it comes to images, video sequences, and similar, it is not only facial features that can identify a person. It may also be possible to identify a person through, for example, body shape, body movements, and gait. The concept of personal data has a broad meaning and includes all kinds of information that can be directly or indirectly linked to a living person.

4.1.2 IMY's Comments

In light of the material (technical descriptions, images, and video clips) presented by the participants in the pilot project, IMY finds that there is much to suggest that there are identifiers in the data generated by the relevant LiDAR sensors and that this data could constitute personal data under Article 4.1 of the General Data Protection Regulation. In a video clip that IMY has reviewed, the data generated by the LiDAR sensor is illustrated. It is clearly possible to discern individuals (see Image 3 above). These individuals could, based on factors such as clothing, body movements, and body shapes, likely be identified by others who are familiar with them in one way or another. Regarding the output of the AI algorithm, it only concerns statistical data on the number of women, men, and children present at the location within a certain time frame. Data that cannot be traced back to a specific individual does not constitute personal data under the General Data Protection Regulation.

23 <https://www.imy.se/verksamhet/dataskydd/det-har-galler-enligt-gdpr/introduktion-till-gdpr/personuppgifter/>

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

17(28)

Date: 2024-02-08

4.2 Processing of Personal Data

4.2.1 What is processing?

Processing of personal data is defined in Article 4.2 of the General Data Protection Regulation (GDPR). Processing essentially refers to anything that can be done with the data. This can include, for example, collecting, recording, storing, analyzing, disclosing, or deleting the data. The concept of processing, like the concept of personal data, has a broad meaning. Almost all handling of personal data is considered processing from a data protection law perspective. It should be noted that the concept of processing is not dependent on how long the personal data processing lasts. This means that even processing that occurs for a very short time constitutes personal data processing according to the GDPR.

4.2.2 IMY's comments

IMY finds that the process that occurs when a LiDAR sensor generates output (i.e., creates the moving 3D point clouds) typically constitutes personal data processing according to the GDPR. Furthermore, there is much to suggest that personal data processing would also occur when an AI algorithm performs calculations based on a LiDAR sensor's output to classify whether an individual within the sensor's detection range is a woman, a man, or a child.

4.3 Sensitive Personal Data

4.3.1 What are sensitive personal data?

Article 9.1 of the GDPR states that processing of special categories of personal data (so-called sensitive personal data) is prohibited. The prohibition does not apply if the processing is supported by one of the exceptions in Article 9.2, for example, if the data subject has given explicit consent to the processing or if the processing is necessary for reasons of substantial public interest based on national law. Sensitive personal data includes, among other things, data about health, sexual orientation, and biometric data used to uniquely identify a person. Biometric data is defined in Article 4.14 of the regulation as "personal data obtained through specific technical processing concerning the physical, physiological, or behavioral characteristics of a natural person, which allows or confirms the identification of that natural person, such as facial images or fingerprint data."

However, according to Recital 51 of the GDPR, the processing of photographs should not systematically be considered as processing of sensitive personal data. Photographs are biometric data only if they are processed with specific technology that enables the identification or authentication of a person.

4.3.2 IMY's comments

IMY has noted that the output generated by a LiDAR sensor could contain biometric data in the form of facial shapes. However, within the framework of this pilot project, IMY has not received any indications that the relevant AI algorithm is capable of doing anything other than classifying whether a certain individual is a woman, a man, or a child. The algorithm should therefore not be able to identify an individual person. In light of this, there is much to suggest that the output generated by the LiDAR sensor does not automatically constitute processing of biometric data.

4.4 Possible avenues for further investigation

Under the conditions that have been relevant in the project, there is much to suggest that the use of LiDAR sensors may involve the processing of personal data. However, this does not mean that the same reasoning is applicable to all forms of uses or measurements conducted with the help of LiDAR sensors or similar technology. The choice of sensor is of great importance. A sensor with such low resolution that individuals cannot be identified falls outside the scope of the GDPR. The same reasoning can be applied if the sensor is positioned at such a distance in relation to the individuals captured by the sensor. The choice of sensor should therefore be carefully examined before use begins. If the same results can be achieved with a sensor of lower resolution and shorter range, such a sensor should be used. The same reasoning can also be applied to edge computing. Processing of data directly in, or in close proximity to, the sensor reduces the risks associated with processing compared to data sent to a central server via an open network.

Another aspect to consider is the LiDAR technology's capacity to process biometric data, which should be taken into account before use begins. This is an additional factor beyond technological development. A low-resolution sensor that today does not have the capacity to identify individual

persons could possibly do so in the near future, considering the increased computational power of processors and refined algorithms.

24 See more in section 3.1 above.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

19(28)

Date: 2024-02-08

5. How does the use of LiDAR sensors relate to the Camera Surveillance Act?

Camera surveillance conducted in Sweden is primarily regulated by the General Data Protection Regulation (GDPR) and the Camera Surveillance Act. The GDPR constitutes the fundamental regulation for the processing of personal data that occurs through camera surveillance, and it includes provisions regarding basic principles, legal basis, information obligations, and the rights of the data subjects.

The Camera Surveillance Act complements the GDPR. It regulates certain issues regarding camera surveillance where the GDPR has left room for national regulations, such as the requirement for permits for certain camera surveillance. It is the IMY that is responsible for deciding on permits for such camera surveillance. However, not everyone conducting camera surveillance needs to apply for a permit.

A permit is required if the camera surveillance involves permanent or regularly repeated monitoring of individuals in a location accessible to the public and provided that the surveillance is to be conducted by an authority or someone other than an authority in the performance of a task of public interest as mandated by law or other regulations, collective agreements, or decisions made under the authority of law or other regulations.

In the current project, the participants wish to use LiDAR sensors, a technology that the IMY has not previously assessed within the framework of the authority's permit or supervisory activities. One of the main questions in the project has been how the relevant technology relates to the Camera Surveillance Act, and more specifically whether the current surveillance may be subject to the permit requirement in the law.

To assess whether surveillance with LiDAR sensors falls under the Camera Surveillance Act and whether the surveillance—under the conditions present in the project—may require a permit or not, several questions need to be answered. Below is an overview of the various factors that need to be considered to determine whether surveillance is permit-required, followed by a general discussion regarding the trade-offs that need to be made when assessing whether the surveillance is permissible or not.

5.1 Does LiDAR technology fall under the concept of "camera surveillance" in the Camera Surveillance Act?

In Section 3 of the Camera Surveillance Act, there is a definition of the concept of camera surveillance. Camera surveillance refers to the use of a television camera, another optical-electronic instrument, or comparable equipment, without being operated on-site, in a manner that entails permanent or regularly repeated monitoring of individuals.

5.1.1 Television camera, other optical-electronic instrument, or comparable equipment

The first question that needs to be answered is whether the technology that the participants wish to use, namely LiDAR sensors, falls under the equipment specified in the Camera Surveillance Act, namely television camera, other optical-electronic instrument, or comparable equipment.

Regarding the equipment covered by the law's definition, the delimitation is not new but was introduced in the law (1990:484) on surveillance cameras, etc. The primary purpose of introducing a new law on surveillance cameras at that time was to adapt

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

20(28)

Date: 2024-02-08

the legal regulation to the technological development. At the time of its introduction, the intention was to have a formulation that was technology-neutral and as unbound by technological development as possible. The formulation was also to encompass technology that could be developed in the future. Thus, the delimitation captures a broad spectrum of technical equipment.

In subsequent laws regarding camera surveillance, the legislator has not deemed it necessary to change which equipment should be included in the material scope of application. One reason for this has been that the delimitation has not led to any particular application difficulties. The delimitation has also been assessed as appropriate to capture technology that does not exist today but may be developed in the future.

In the preparatory works, the meanings of the respective concepts have been developed:

- Surveillance with TV cameras was, at the time of the 1990 law on surveillance cameras, the most commonly used equipment for personal surveillance. The fact that the term still exists in the currently applicable camera surveillance law is due to it being considered an established term that is often used in signage regarding camera surveillance.

- The term optical-electronic instrument includes two terms that each have a specific meaning. The term optical refers to all recording that can occur with instruments within the electromagnetic wavelength spectrum for optical radiation. The expression electronic means that the transmission of images, the display of images on a monitor, or the storage of images occurs through electronic influence. In other words, the term optical-electronic instrument refers to equipment with which an image capture can be transmitted electronically to a display screen or stored in a computer memory or on another electronic medium.

- The expression comparable equipment refers to, for example, instruments that can utilize electromagnetic radiation such as X-rays and radiofrequency radiation.

5.1.2 IMY's Comments

As indicated by the technical description of LiDAR sensors in section 3.1 above, a sensor calculates the distance to an object by emitting infrared laser pulses and then measures the time it takes for the pulses to be reflected back to the light receiver in the sensor. The LiDAR sensor then generates output in the form of a moving point cloud in 3D format, which in the current project is to be transmitted through electronic influence.

A LiDAR sensor uses wavelengths in the spectrum around 900 nm. IMY notes that the laser pulses emitted by a LiDAR sensor can be classified within the electromagnetic wavelength spectrum for optical radiation. Optical radiation typically refers to the subset of the electromagnetic spectrum that ranges from a wavelength of approximately 100 nm up to about 1 mm, which includes ultraviolet radiation (UV), visible radiation (light), and infrared radiation (heat). In this context, and in

25 Prop. 1989/90:119 p. 1.

26 SOU 2017:55 p. 210.

27 Prop. 1989/90:119 p. 39 ff.

28 SOU 2017:55 p. 210.

29 See SOU 2018:92 p. 61 for an overview of the wavelengths included in the electromagnetic wavelength spectrum for optical radiation.

30 Prop. 1989/90:119 p. 39.

31 SP Report 2013:36 p. 7. Artificial optical radiation in Swedish industry (<https://www.diva-portal.org/smash/get/diva2:962758/FULLTEXT01.pdf>).

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

21(28)

Date: 2024-02-08

in accordance with what is stated in the preparatory works regarding the equipment that the Camera Surveillance Act is to encompass, IMY finds that LiDAR sensors typically fall under the concept of other optical-electronic instruments in the Camera Surveillance Act.

5.2 Is it a matter of permanent or regularly repeated personal surveillance?

For surveillance to fall within the material scope of the Camera Surveillance Act, it is not sufficient that the technical equipment is covered by the Camera Surveillance Act. An additional requirement is that the technology is used in a manner that entails permanent or regularly repeated personal surveillance.

5.2.1 Permanent or regularly repeated personal surveillance

The concept of personal surveillance is a narrower concept than the concept of personal data (see section 4.1.1 above). Personal surveillance means that individuals can be identified through the surveillance. According to the preparatory works of the Camera Surveillance Act, it is required that one can distinguish the observed individuals from other individuals without significant uncertainty for the possibility of identification to be considered present. This is the case, for example, if the entire person or the person's face is clearly visible. Even distinctive clothing, special body movements, or particular body constitution can enable identification.³² This means that camera surveillance used solely to read vehicle registration numbers, but which does not film people, is not considered personal surveillance. However, it may constitute processing of personal data under the General Data Protection Regulation.

For the Camera Surveillance Act to be applicable, it is also required that the personal surveillance is permanent or regularly repeated. Permanent personal surveillance refers, for example, to surveillance over an extended period of a location where people typically gather. This may involve surveillance in a square, on a public transport vehicle, in a waiting room, or in a store.

Regularly repeated personal surveillance refers to personal surveillance that occurs on multiple occasions and with a certain regularity. Surveillance that occurs at times spread over time can be considered regular if it involves a systematic use of surveillance equipment.³³ This means that, for example, cameras mounted next to a walking path in the forest may fall within the scope of the Camera Surveillance Act if people pass along the path with a certain regularity, even if it does not occur to a large extent.

The purpose of the use of the camera equipment is irrelevant to the question of the applicability of the Camera Surveillance Act.³⁴ Even if the purpose of the surveillance is not to monitor people, the law may still become applicable if people enter the camera's field of view for an extended period, or at least somewhat regularly, making it possible to identify them directly or indirectly.

5.2.2 IMY's comments

In the project, IMY has been provided with, among other things, images and video clips showing sequences of output from a LiDAR sensor with a resolution corresponding to what the participants wish to use. In the images (see image 3 in section 3 above) and video clips that IMY has reviewed, the LiDAR sensors have, in some cases, been mounted on a moving vehicle and in other cases fixed at a height of 4–8 meters above the ground. The individuals captured by the sensor have been at various distances of 5–30 meters from the sensor. The participants in

³² Prop. 2017/18:231 p. 41 and 137.

³³ Prop. 2017/18:231 p. 137.

³⁴ Prop. 2017/18:231 p. 137.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

22(28)

Date: 2024-02-08

The project has stated that it will be possible to access output directly from the LiDAR sensor before processing occurs using the AI algorithm, for example during calibration and installation. These factors need to be considered when assessing whether individuals can be identifiable.

In comparison with traditional surveillance cameras, IMY initially notes that the output generated by a typical LiDAR sensor makes it more difficult for the human eye to identify a person's face, especially if the person is located a bit further away from the sensor. It may thus be challenging to distinguish one person from another. However, it is important to note that the output generated by a typical LiDAR

sensor consists of 3D point clouds, meaning that the images contain depth data, which may allow for the creation of precise 3D models of individuals within the sensor's detection range.

Based on the conditions of the project, IMY concludes that it may be difficult to identify individuals by distinguishing their faces in the output generated by the LiDAR sensor, particularly since the surveillance is intended to occur at a relatively long distance. However, IMY finds that, based on the material it has reviewed, it may be possible to discern and identify individuals based on, for example, body movements, body constitutions, and distinctive clothing. Therefore, IMY finds that there is much to suggest that the output generated by the LiDAR sensors constitutes personal surveillance.

At the location where the LiDAR sensors are intended to be used, namely Jan Stenbeck Square in Kista, a large number of people move about. The Traffic Office's wish is to conduct measurements at the square during certain time intervals between 07:00 and 22:00 two days per month. IMY notes that the measurements will thus occur regularly. It therefore seems likely that this could constitute a form of repeated personal surveillance that falls under the scope of the Camera Surveillance Act.

5.2.3 Possible avenues for further investigation

Under the conditions that have been relevant in the project, IMY has concluded that the use of LiDAR sensors may involve personal surveillance. However, this does not mean that the same assessment is valid for all forms of surveillance or measurements using LiDAR sensors or equivalent technology.

How the surveillance is organized can affect the assessment of whether it constitutes personal surveillance or not. One factor that may influence this is the placement of the LiDAR sensor in relation to distance and angle to the objects that are hit by the sensor's laser pulses. Since identification of individuals is primarily assessed to be possible based on body movements and body constitution, and considering that the output from the LiDAR sensor has lower resolution the further away the object is, it may be difficult to identify individuals through the LiDAR sensor's output at a certain distance.³⁵ IMY assesses that this is a significant difference compared to traditional surveillance cameras, which often allow for the identification of individuals even at greater distances. Another factor that influences this is the LiDAR sensor's resolution, field of view, and range (see more in sections 3.1 and 4.4 above).

³⁵ It should be noted, however, that the purpose of the intended use of the LiDAR sensors in the current pilot project is to identify whether a person is a woman, a man, or a child. To achieve this purpose, it will likely require relatively clear data of the individuals captured by a LiDAR sensor.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

23(28)

Date: 2024-02-08

5.3 Permit-required surveillance – What considerations need to be made and how is the assessment affected by new technology?

5.3.1 Is the use of LiDAR sensors in a public place permit-required?

According to Section 7 of the Camera Surveillance Act, a permit is required for camera surveillance of a location accessible to the public if the surveillance is to be conducted by an authority or someone other than an authority in the performance of a task of public interest as mandated by law or other regulations, collective agreements, or decisions made under the authority of law or other regulations. Authorities refer to all state and municipal bodies, with the exception of the Parliament and municipal and regional councils.

The permit requirement applies to all authorities regardless of the type of activity in which the camera surveillance is used and regardless of the purpose of the surveillance.³⁶ It is IMY that is responsible for permit decisions under the Camera Surveillance Act.

The Traffic Committee of the City of Stockholm would be the data controller for the intended surveillance using LiDAR sensors. The Traffic Committee is an authority in the sense of the Camera Surveillance Act. Furthermore, the LiDAR sensors are intended to be used in a location accessible to the public. Under such circumstances, it is likely that the surveillance would be assessed as permit-required.

5.3.2 Purposes that should be specifically considered in a permit assessment

The purpose of the second legal question in the project has been to assess how the use of LiDAR technology relates to the Camera Surveillance Act. It is important to emphasize that IMY can only provide guidance within the framework of the regulatory sandbox and cannot give binding statements on whether a specific processing of personal data through camera surveillance will be permitted or not. However, there is reason to generally discuss the considerations that must be made according to the Camera Surveillance Act based on the purposes and the technology relevant to the pilot project. A permit for camera surveillance should be granted if the interest in such surveillance outweighs the individual's interest in not being surveilled. This is stated in Section 8, first paragraph of the Camera Surveillance Act. The second paragraph of the same provision states that in assessing the interest in camera surveillance, it should be particularly considered whether the surveillance is necessary for any of the specific purposes outlined in the provision. The purposes that should be specifically considered are whether surveillance is needed to:

1. prevent, hinder, or detect criminal activity or investigate or prosecute crimes in a crime-prone location or in another location where there is a particular risk of attacks on someone's life, health, or safety or on property,
2. prevent, hinder, or detect disturbances of public order and safety or limit the effects of such disturbances,
3. exercise control activities,
4. prevent, hinder, or detect accidents or limit the effects of accidents that have occurred, or
5. serve other comparable purposes.

If the surveillance is conducted for any of these purposes, it means that the interest in surveillance is considered to weigh heavier than otherwise.

In assessing the individual's interest in not being surveilled, it should be particularly considered how the surveillance will be carried out, whether technology that promotes the protection of the individual's

36 Prop. 2017/18:231 p. 141.

Swedish Authority for Privacy Protection

Diary Number: IMY-2023-15495

24(28)

Date: 2024-02-08

Personal integrity shall be used and which area shall be monitored (Section 8, third paragraph of the Camera Surveillance Act).

5.3.3 IMY's Comments

It can be noted that the purposes that should be particularly considered during the permit assessment are purposes that are traditionally associated with camera surveillance. The majority of the permit applications that IMY receives also pertain to monitoring for such purposes. This concerns, for example, camera surveillance to prevent crimes and accidents. However, with new, better, and cheaper technology, camera surveillance for other, less traditional purposes has become more common. For instance, cameras are now used to control traffic lights and measure traffic flows. This project concerns the Traffic Office's desire to use LiDAR sensors to produce statistics on the number of women, men, and children moving in Jan Stenbeck Square in Kista. With the help of this statistics, the Traffic Office can assess what measures need to be taken to create public spaces that are perceived as attractive, vibrant, and safe. It is therefore not given that the intended purpose of the monitoring falls under the purposes that should be particularly considered according to Section 8, second paragraph of the Camera Surveillance Act.

Even if the camera surveillance is not to be conducted for such a purpose, it may still be possible to obtain permission for the monitoring. However, this requires that the purpose is justified and that the interest in camera surveillance outweighs the individual's interest in not being monitored. More in-depth considerations may also be needed regarding the necessity of camera surveillance in relation to privacy risks and how these risks can be minimized.³⁷

The measurements shall take place in a square adjacent to a shopping center, a location where a large number of people may be present at times. Some will only pass through the square, but there are

also benches and other seating areas as well as outdoor cafés where people may stay for a longer period. It is thus a matter of a location where the interest in privacy weighs heavily, as a large number of people may be subject to monitoring. For monitoring at such a location to be permitted, either the interest in monitoring must weigh heavily or the privacy risks associated with the monitoring must be minimized.

As previously noted, LiDAR sensors generate output where it is typically more difficult to identify individuals with the naked eye compared to traditional camera surveillance. In the project, data processing will also take place directly in, or in close proximity to, the sensor (see section 3.1.1 above). Output from the sensor, where identification of individuals may be possible, shall be deleted once the data has been processed in the AI algorithm. Output from the AI algorithm shall only consist of data regarding the number of women, men, and children present at the relevant location, that is, only statistical data that does not contain personal data.

The fact that personal data will only be processed for a short time and then deleted is a factor that may influence the assessment of the privacy interest; a shorter time generally reduces privacy risks. However, the data will need to be retained for at least a certain period to avoid double counting when a person remains at the location. The processing being conducted directly in, or in close proximity to, the LiDAR sensor also means that the risks of privacy infringement are reduced compared to if personal data had instead been sent via, for example, the internet to a central server.

37 Prop. 2017/18:231 p. 144.
Integrity Protection Authority
Diary Number: IMY-2023-15495
25(28)

Date: 2024-02-08

The output from the LiDAR sensor will only be reviewed during calibration and installation, and not during normal operation, which may also be a factor influencing the risks of privacy infringements. Overall, IMY can conclude that the chosen technology can help to limit privacy infringements, especially in comparison with traditional surveillance cameras. As IMY has noted in section 4.3 above, the output generated by a LiDAR sensor could contain biometric data in the form of facial shapes. If such data were to be used in surveillance to, for example, identify individual persons, it could involve the processing of sensitive personal data. This is an aspect that increases privacy risks and that the user of the technology should consider before commencing surveillance. For the processing of sensitive personal data to be permissible, there must be support for the processing in one of the exceptions in Article 9.2 of the General Data Protection Regulation. Examples of exceptions include if the data subject has given explicit consent or if the processing is necessary for reasons of substantial public interest based on national law. However, it should be emphasized that the AI algorithm discussed within the framework of this pilot project should not be able to identify individual persons.

5.4 The Camera Surveillance Act is Under Review

In March 2023, the government initiated a camera surveillance investigation aimed at proposing measures to facilitate camera surveillance for municipalities, regions, and other entities than authorities that perform a task of public interest. The assignment includes, among other things, mapping the need for simplified rules regarding camera surveillance by considering the removal of the permit requirement for, among others, authorities under the Camera Surveillance Act. Furthermore, the need for expanded opportunities for camera surveillance to combat crime and maintain order and security will be analyzed. The investigation is to be completed no later than April 15, 2024. When this final report is published, it is still unclear what conclusions or proposals for action the investigation will arrive at.

On December 21, 2023, the investigation received an additional directive that expands the assignment to also include proposals to facilitate the Police Authority's handling prior to the commencement of camera surveillance and proposals that would exempt the authority in more cases from the requirement for notification and the right to information in camera surveillance.

38 2023 Camera Surveillance Investigation. Easing Restrictions for Camera Surveillance in Certain

Activities, dir. 2023:40.

39 Additional Directive to the 2023 Camera Surveillance Investigation, dir. 2023:182.

Swedish Authority for Privacy Protection

Case Number: IMY-2023-15495

26(28)

Date: 2024-02-08

6. Reflections on Regulatory Sandbox as a Working Method

IMY has now conducted two pilot projects in the regulatory sandbox. In the autumn of 2023, it was decided that the regulatory sandbox will become a permanent part of IMY's operations. Two new projects are to be carried out in 2024. According to IMY's experiences, this working method creates benefits and learning opportunities for both the relevant businesses and supervisory authorities. Since IMY began operating its regulatory sandbox, many public entities have reached out to the authority to learn more about the working method. IMY has, among other things, lectured and engaged in dialogue with the Energy Markets Inspectorate, eSAM, the Swedish Civil Contingencies Agency (MSB), the Swedish eHealth Agency, the Swedish Transport Agency, and Vinnova. Furthermore, IMY holds regular meetings with other European data protection authorities that operate regulatory sandboxes to discuss ongoing projects and exchange experiences.

According to the forthcoming AI Regulation, member states are to establish regulatory sandboxes for AI. To the extent that an AI system, which is subject to the regulatory sandbox for AI, involves the processing of personal data, the national data protection authority will be linked to the operation. It is therefore likely that IMY, as the national data protection authority, will be involved in several such projects.

Although the regulatory sandboxes governed by the forthcoming AI Regulation may not necessarily be operated in the same way as IMY's regulatory sandbox, IMY has now gained valuable knowledge on how to conduct projects in new, exploratory ways.

IMY sees that the regulatory sandbox creates benefits not only for ourselves but, above all, for innovators, for authorities that interpret and apply regulations, and for society at large.

For innovators, the benefit is that they receive better, more, and earlier guidance and support regarding how data protection regulations should be interpreted and applied. Time can be saved and risks reduced, which means that businesses can more quickly and safely take advantage of modern technology. By documenting and sharing reasoning and assessments, more than just those who participated in a project in the regulatory sandbox can receive guidance and benefit from the lessons learned from the work.

For IMY, the significant added value is an increased knowledge of and understanding of innovative technology and the legal issues and challenges that arise in its application. For IMY, this pilot project has contributed to a deeper understanding of how LiDAR technology works. It has also provided important insights into how we, as a supervisory authority, can develop and improve our work in providing guidance and support to innovation actors.

At the societal level, the regulatory sandbox is assessed to be a concrete way to promote innovation and the development of new technology while also contributing to ensuring data protection and information security. The working method is not the sole solution, but it can be a piece of the puzzle to improve Sweden's competitiveness, increase the pace of our digital transformation, and contribute to sustainable technological solutions that citizens can trust.

Technological development is advancing rapidly, and legislation often struggles to keep up with developments. Work in regulatory sandboxes can lead to the identification of the need to change existing regulations more quickly and to bring this to the attention of lawmakers at an earlier stage. In this way, regulatory sandboxes can create better conditions for legislative work to keep pace with technological development.

One risk that IMY has identified with the concept of a regulatory sandbox is that some businesses may have misunderstandings or unrealistic expectations about what this working method entails. This has not been the case in this pilot project, but in dialogue with others, we have sometimes encountered the incorrect perception that IMY's regulatory sandbox is associated with exemptions from applicable

rules.

There may also be a risk that businesses interested in participating in a project in the regulatory sandbox have an expectation that participation will always result in the desired processing of personal data being able to be carried out in some way. This is not guaranteed, as IMY could also conclude that it is questionable whether there are legal possibilities to carry out the intended processing of personal data. It is also important to emphasize that IMY can only provide guidance within the framework of the regulatory sandbox and not binding decisions on whether a specific processing of personal data will be permitted or not.

For IMY, and other authorities considering operating their own regulatory sandboxes, it is therefore an important task to clearly communicate what the working method entails and what participants can expect.

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7. Reflections on Applying Data Protection Regulations to New Technology

Based on the work in the pilot project, some general reflections can be made about practically working with innovation and new technology while ensuring good data protection. These reflections are deemed relevant even when the innovation process occurs within the same organization.

- A cross-functional approach, where legal and technical expertise closely collaborates with representatives of the core business, is crucial for success.
- A relatively deep understanding of the technology is necessary to make relevant legal assessments. It is often essential for lawyers to gain an understanding of how the relevant technology works early in the process in order to assess fundamental data protection legal issues, for example.
- Just as technicians need to educate lawyers on how the technology works, lawyers need to have good pedagogical skills to explain the data protection regulations and how they are structured. In order to embed good data protection, those developing the technology may need to understand the fundamental principles of data protection regulations, that every processing of personal data needs to be supported by a legal basis, to have built-in data protection as a starting point, etc. Therefore, a skilled data protection lawyer needs to develop good pedagogical abilities.
- In addition to lawyers needing to understand the technology and technicians needing to understand the law, structures and tools are needed to ensure a common understanding within the working group. Simple but concrete methods may include jointly identifying and defining key concepts in a glossary, concluding meetings with a summary, keeping joint minutes, and ensuring time for clarification questions.
- The overarching legal questions can likely be identified when the work begins. However, a central part of the legal work is to continuously revisit and, if necessary, adjust the legal questions that need to be investigated. Are the questions initially posed still relevant? Which provisions of the data protection regulation may be applicable? Is there other legislation that also needs to be considered?
- It may be a good idea to continuously consider whether there are additional internal or external stakeholders that need to be involved in the process. This may involve other expert authorities, but also internal functions such as archives, procurement, purchasing, or security. It may also be relevant to seek support, guidance, or inspiration from experts in a specific technology area.